

How do World Models and Policies Compose in LLM Agents? A Joint Spectral and Behavioral Account

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Abstract

How do LLM agents come to both understand environments they act in and master tasks set within them? Through controlled experiments combining world-model training (next-state prediction) and policy training (reward maximization), we investigate this question. We dissect the resulting models through their additive parameter updates. Geometrically, we find effective world-model updates are low-rank and share an *input*-feature subspace with policy updates while writing to nearly orthogonal *output* directions, whether trained separately or sequentially. However, we find that, in projection interventions, the sequential update induces more robustness than separate policy RL when removing the world model’s leading input directions, suggesting that it has learned alternative input pathways. Behaviorally, we find the sequentially trained agent explores a wider range of states and actions. Based on this, we ask: does policy training preserve world knowledge as well as it *could*? We probe this with training-free merging built on the geometrically motivated input basis plus an online world-model loss during policy RL, and show both improve over the untreated baseline. Our findings suggest world knowledge and task-directed ability can be learned in geometrically complementary forms, and that future post-training pipelines should consider how best to engineer the interface between them.

1 Introduction

We know things about the world, and we do things in it. Language models too are increasingly studied through this dual lens (e.g., their *world models* (Li et al., 2023; Toshniwal et al., 2022; Guan et al., 2023; Vafa et al., 2024)) and are even explicitly trained to internalize both these capacities. For example, post-training pipelines for LLM-based agents increasingly adopt a two-stage formula: a *world-model* stage that trains the model

to anticipate environment dynamics, followed by a *policy* stage that optimizes end-task reward (Yu et al., 2025b; Zhang et al., 2026; Yu et al., 2025a; Chen et al., 2025; Yu et al., 2026). This sequential pipeline consistently improves over policy training from a base model, often by sizable margins.

What is the relationship between these two objects: knowledge and ability? Intuitively, knowing your environment well helps you select better actions. Still, the two are not the same: by analogy, radiologists don’t necessarily make good surgeons. Despite the empirical success of world-model learning, the precise mechanism by which it benefits downstream policies remains opaque. For example, do world models act as a warm-up that places the policy at a better initialization in a region of weight space it would have reached on its own? Does it identify a useful subspace the policy stage then refines? Or does it leave behind a representational structure on which the policy builds in meaningfully *complementary* directions?

Getting good answers to these questions is potentially consequential for both interpretation and design. If world-model learning accelerates the policy’s discovery of the same solution, the second stage should in principle be able to absorb the first, and the pipeline could be collapsed without loss. If instead the two stages contribute structurally different things, then how to preserve, combine, and schedule them is an important design lever for agentic post-training. Prior work has documented the benefit of the pipeline but rarely its mechanism. Existing analyses either evaluate the composed pipeline as a black box (Yu et al., 2026; Chen et al., 2025) or characterize properties of post-training updates in isolation (Aghajanyan et al., 2021; Hu et al., 2022; Zhu et al., 2025), without directly contrasting world-model and policy stages. Perhaps closest to the present work is Zhu et al. (2025), which uses spectral structure to show how RLVR largely preserves principal subspaces and updates

Code: github.com/Rick-Xu315/WM_Spectral

off-principal directions vs. SFT. Similarly, we analyze post-training updates through spectral geometry and, consistent with their results, we find RL and SFT updates differ sharply in the world model regime. However, our primary focus is within the RL regime, analyzing the input-feature vs. output directions.

We take a first step toward understanding how world-model and policy stages interact via parameter-space dissection of a controlled family of post-trained checkpoints. For each post-training paradigm X , we form the additive parameter update $\Delta_X = \theta_X - \theta_0$ relative to the base model; the policy stage’s incremental contribution on top of the world model can then be analyzed separately. We then conduct a series of spectral interventions (subspace projection, rank truncation, and magnitude-based pruning) and behavioral measurements (held-out next-state prediction entropy, state coverage, and patterns of success). Our results show meaningful complementarity between world-model and policy updates: the two stages share an input-feature subspace but write to nearly orthogonal output directions, and sequential training additionally equips the policy update with input-pathway redundancy that base-initialized policy training cannot acquire. We also show that the two-stage agent exhibits useful behavioral capacities (e.g., preserved world-model calibration, expanded state coverage) over the policy-only version.

In all, this paper contributes an exploratory empirical analysis of two-stage agentic post-training: a parameter-space dissection that maps out the geometric structure underlying the two-stage recipe’s empirical success, across two benchmarks and two models, in which we ask:

- RQ1** Are there geometric and behavioral signatures of effective world-model training?
- RQ2** What is the relationship between parameter updates produced by world-model and policy training, independently vs. sequentially?
- RQ3** Is the world-model preserved enough through policy training, and can we improve this?

This work then contributes:

1. **Empirical evidence** showing **spectral and behavioral signatures of effective world-model training** (RQ1; §4, §6).
2. A **Δ -decomposition** and **left/right subspace projection methodology**, using which we

find **geometric evidence of stage complementarity** (RQ2; §5). We find that RL-trained world models’ left singular subspaces (output directions) are nearly orthogonal to the downstream policy’s, while their right singular subspaces (input-feature directions) are strongly aligned. Sequential training additionally equips the policy update with redundant input pathways unavailable from base-only.

3. **Design implications** for *preserving* world models during policy training, realized via **merging and SFT experiments** (RQ3; §7). The complementarity finding predicts that the two stages should compose constructively. We test both a training-free KnOTS/TIES merge and an online world model SFT auxiliary loss during policy learning (similar to concurrent work by Shrivastava et al. (2026)), both of which improve over the untreated baseline.

2 Preliminaries

2.1 Problem Setup

We formulate multi-step task completion as a Markov Decision Process $\langle \mathcal{S}, \mathcal{A}, \mathcal{T}, \mathcal{R}, \gamma \rangle$ (Sutton and Barto, 2018). A ReAct-style (Yao et al., 2023) LLM agent π_θ interleaves reasoning and acting: at step t , it receives an observation $s_t \in \mathcal{S}$ together with up to H turns of interaction history $\langle s_{t-H}, a_{t-H}, \dots, s_t \rangle$, produces thought τ_t , and emits an action $a_t \sim \pi_\theta(\cdot \mid s_{t-H:t}, a_{t-H:t-1})$; the environment returns the next observation $s_{t+1} \sim \mathcal{T}(\cdot \mid s_t, a_t)$. The loop terminates with a task-success reward r_T .

World model. A world model $\mathcal{W}_\phi$ is a learned surrogate for the transition $\mathcal{T}$ (Ha and Schmidhuber, 2018; Hafner et al., 2023): it predicts the next state $\hat{s}_{t+1} \sim \mathcal{W}_\phi(\cdot \mid s_{t-H:t}, a_{t-H:t})$. We write $\hat{s}_t$ for predicted and s_t for ground-truth states.

2.2 Two-stage Training

A growing line of work (Yu et al., 2025b; Zhang et al., 2026; Yu et al., 2025a, 2026) adopts a two-stage post-training pipeline for agentic LLMs: a *world-model* stage that grounds π_θ in environment dynamics, followed by a *policy RL* stage that optimizes task-success reward.

World-model SFT. The simplest instantiation finetunes π_θ on transition triplets to maximize $\log \pi_\theta(s_{t+1} \mid s_{t-H:t}, a_t)$ (Yu et al., 2025b; Zhang

et al., 2026; Yu et al., 2025a; Chen et al., 2025). WMSFT inherits the pitfalls of imitation learning: token-level rather than semantic supervision, sensitivity to data quality, and model collapse / catastrophic forgetting (Yu et al., 2026).

World-model RL (WMRL). Yu et al. (2026) train the world model with RL on a self-supervised *sim-to-real* reward, which we call WMRL. Given a prediction $(\tau_t, \hat{s}_{t+1}) \sim \pi_\theta(\cdot \mid s_{t-H:t}, a_t)$, the reward

$$r^{\text{WM}} = \mathbf{1}[\text{sim}(E(\hat{s}_{t+1}), E(s_{t+1})) > \eta] \quad (1)$$

scores embedding-space proximity to the realized next state, where $E(\cdot)$ is a frozen embedding model, $\text{sim}(\cdot, \cdot)$ a similarity function, and η a threshold. π_θ is optimized with GRPO (Shao et al., 2024) on its own rollouts using this reward, thereby encouraging *semantic* next-state alignment. We test $\eta \in \{0.5, 0.8, 0.9\}$, and also test alternates including LLM-as-judge evaluation (no- η) and world-model transitions extracted from official ALFWorld expert data.

Policy RL (PRL). The second stage then optimizes π_θ end-to-end for task success with GRPO:

$$\mathcal{L}_{\text{GRPO}} = -\mathbb{E}_{\pi_{\theta_{\text{old}}}} \left[\min(\rho_\theta A, \text{clip}(\rho_\theta, 1 \pm \epsilon) A) - \beta D_{\text{KL}}(\pi_\theta \parallel \pi_{\theta_{\text{ref}}}) \right],$$

with $\rho_\theta = \pi_\theta(y \mid x) / \pi_{\theta_{\text{old}}}(y \mid x)$ and group-relative advantage $A = (r_T - \mu_G) / \sigma_G$.

2.3 Benchmarks and Models

Our primary benchmark is **ALFWorld** (Shridhar et al., 2021), a text-based embodied environment in which the agent locates and interacts with household objects to complete natural-language tasks. It tests long-horizon planning and tracking of action-conditioned state transitions. We use τ^2 -**Bench** (Barres et al., 2026) as a secondary test, which is an interleaved tool-use environment in which the agent alternates between tool calls and dialogue with a simulated user.

Model preparation. We follow the training setup of Yu et al. (2026): Qwen2.5-7B-Instruct (Qwen Team, 2025a) for ALFWorld (Qwen3-8B (Qwen Team, 2025b) for τ^2 -Bench). The checkpoints studied throughout the paper are re-trained in this setup: the base model, WMRL-only, PRL-only, and sequential WMRL $\rightarrow$ PRL. Full data, hyperparameter, and prompting details are in Appendix A.1 and B.1.

Replicated performance. Table 1 reports ALFWorld results; our numbers track Yu et al. (2026). See Appendix B.1 for τ^2 -Bench replication.

Table 1: Replicated success rates on ALFWorld (%), mean $\pm$ std over 3 seeds. Our numbers closely match Yu et al. (2026).

Model	ID	OOD	AVG
ReAct (base)	19.8 $\pm$ 4.5	15.6 $\pm$ 1.3	17.7 $\pm$ 2.9
WMSFT	5.0 $\pm$ 1.0	2.1 $\pm$ 1.5	3.5 $\pm$ 1.2
WMRL	33.6 $\pm$ 3.4	31.8 $\pm$ 5.3	32.7 $\pm$ 3.3
PRL	84.6 $\pm$ 1.3	71.9 $\pm$ 2.6	78.3 $\pm$ 0.9
WMSFT $\rightarrow$ PRL	66.4 $\pm$ 1.3	55.2 $\pm$ 2.0	60.8 $\pm$ 1.5
WMRL $\rightarrow$ PRL	87.0 $\pm$ 0.4	81.8 $\pm$ 0.7	84.4 $\pm$ 0.3

3 Methodology

We first define several measures we use to study the checkpoints mentioned.

Δ -decomposition. Let θ_0 denote the base-model parameters. Following the task-vector formulation of Ilharco et al. (2023), for each post-training paradigm $X \in \{\text{WMSFT}, \text{WMRL}, \text{PRL}, \text{WMRL} \rightarrow \text{PRL}\}$ we obtain trained parameters θ_X and define the additive update $\Delta_X = \theta_X - \theta_0$. This yields four primary deltas: Δ_{WMSFT} , Δ_{WMRL} , Δ_{PRL} (PRL trained directly from base), and $\Delta_{\text{WMRL}+\text{PRL}}$ (the cumulative update after sequential WMRL $\rightarrow$ PRL). To isolate the policy stage’s post-WMRL (conditional) contribution, we further define

$$\Delta\Delta_{\text{PRL}} = \Delta_{\text{WMRL}+\text{PRL}} - \Delta_{\text{WMRL}} \quad (2)$$

so that $(\Delta_{\text{WMRL}}, \Delta\Delta_{\text{PRL}})$ cleanly decomposes the sequential pipeline. All deltas are computed module-wise on the seven trainable linear projections of every transformer block ($\{\text{q}, \text{k}, \text{v}, \text{o}, \text{up}, \text{gate}, \text{down}\}_{\text{proj}}$). Since all checkpoints in our setting share the same architecture and pretrained initialization, their updates also share a common coordinate system. This facilitates a more principled weight-space comparison than for independently trained networks where parameter similarity can be distinct from functional similarity. Additionally, prior work has shown that for models fine-tuned from a common pretrained base, linear operations on weights can correspond predictably to linear changes in intermediate features (Zhou et al., 2024), which offers further empirical support for treating their parameter updates as comparable objects.

Spectral interventions. For each module $\Delta \in \mathbb{R}^{m \times n}$ we compute the SVD $\Delta = U\Sigma V^\top$ and then compute derived quantities defined below.

Effective rank (Roy and Vetterli, 2007), how broadly Δ spreads across singular directions:

$$\text{eRank}(\Delta) = \exp\left(-\sum_i p_i \log p_i\right), \quad p_i = \sigma_i / \sum_j \sigma_j,$$

where σ_i are singular values of Δ and p_i is the spectrum-normalized mass.

Truncated rank- k approximation. The Eckart–Young best rank- k approximation (Eckart and Young, 1936) $\Delta_k = U_{:,1:k} \Sigma_{1:k,1:k} V_{:,1:k}^\top$ gives a coordinate-free notion of how much of an update’s information lives in its top- k singular components.

Subspace projection. For two updates Δ_A, Δ_B at the same module, let $U_A^{(k)}$ and $V_A^{(k)}$ denote the leading k left and right singular vectors of Δ_A . Taking the left projection as an example, we construct three projected variants of Δ_B : parallel, $U_A^{(k)} U_A^{(k)\top} \Delta_B$; orthogonal, $(I - U_A^{(k)} U_A^{(k)\top}) \Delta_B$; and random, $R^{(k)} R^{(k)\top} \Delta_B$, where $R^{(k)}$ is a random orthonormal k -frame of matching dimension.

The reconstructed checkpoint $\theta_0 + \Delta_{\text{WMRL}} + \text{proj}(\Delta \Delta_{\text{PRL}})$ (or $+\text{proj}(\Delta_{\text{PRL}})$ in the independent control) is then re-evaluated end-to-end.

Module-wise subspace overlap. For two leading- k subspaces with orthonormal bases B_A, B_B , the mean squared principal cosine $\frac{1}{k} \sum_i \sigma_i^2(B_A^\top B_B)$ measures alignment (random baseline $\approx k/d$). This is a standard principal-angle-based subspace alignment measure (Björck and Golub, 1973).

Magnitude pruning of Δ . We also retain only the top- $P\%$ of Δ entries by absolute magnitude (zeroing the rest), following task-vector merging conventions (Yadav et al., 2023; Yu et al., 2024). This complements the spectral analysis by operating in coordinate rather than singular-vector space.

Behavioral readouts. We complement the geometric analyses with per-token entropy of the model’s distribution over the *ground-truth* next-state tokens on a held-out WMRL evaluation set (lower = better world-model calibration), and pairwise success-set overlap on the downstream task to flag behavioral complementarity.

4 What makes a good world model? Spectral signatures of effective two-stage training (RQ1)

Figure 1 plots effective rank against success across WM-stage checkpoints, observing that **within RL**

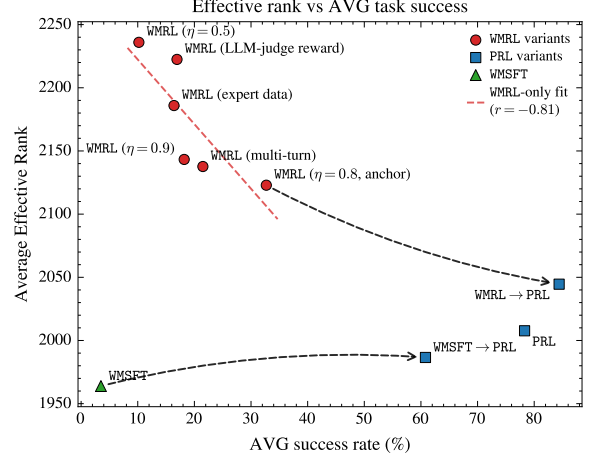


Figure 1: Effective rank vs. AVG success across WM-stage checkpoints: a WMSFT baseline and a WMRL sweep over reward threshold, judge type, data source, and prediction format (Appendix A.2). Within WMRL, lower effective rank tracks higher performance; WMSFT collapses despite low rank.

regime, the strongest world model is low-rank. However, WMSFT operates in a different regime.

WMSFT achieves lowest eRank of any checkpoint yet performs catastrophically. WMSFT $\rightarrow$ PRL trails WMRL $\rightarrow$ PRL by >20 AVG points. Thus, the two regimes are not comparable spectrally; one possible explanation is that WMSFT concentrates updates on surface textual patterns, vs. WMRL looking for embedding-level semantic match and learning dynamics-relevant directions. We defer testing this hypothesis systematically to future work, but note that this split echoes evidence that effective fine-tuning updates concentrate in low-dimensional subspaces (Aghajanyan et al., 2021; Hu et al., 2022) and that RL post-training has a characteristic spectral structure distinct from SFT (Zhu et al., 2025).

WMRL initialization broadens the directions policy RL explores. Comparing Δ_{PRL} against $\Delta_{\text{WMRL}+\text{PRL}}$, the WMRL-initialized pipeline shows substantially higher per-layer effective rank at every depth, with the largest gap in the early-to-mid layers. We posit that the world-model stage opens up directions in weight space that the policy stage subsequently exploits—a geometric counterpart to the behavioral claim that WMRL contributes complementary structure rather than redundant capacity.

The world-model update is functionally low-rank. The low effective rank of strong WMRL checkpoints has a functional counterpart: truncating Δ_{WMRL} to its top singular components while leaving $\Delta \Delta_{\text{PRL}}$ intact incurs little performance cost.

Even a rank-1 truncation (a single outer product) recovers within ≈ 10 points of the full sequential pipeline, whereas the same truncation applied to $\Delta\Delta_{\text{PRL}}$ or to Δ_{PRL} collapses performance to near the WMRL-only model at small k . The world-model stage’s downstream value is therefore concentrated in a handful of directions, while the policy update spreads its task value more diffusely. We refer the reader to Appendix A.5 for the full results.

5 Weight-space Relationship between Δ_{WMRL} and the Policy Stage (RQ2)

Since WMRL and PRL leave deltas of different rank, we ask whether the directions populated by the two stages overlap or are disjoint. Is their relationship more one of *complementarity* or of *redundancy*? We identify two accounts that could explain the observed two-stage WMRL $\rightarrow$ PRL performance gain:

Account A (shared / parallel). The policy stage refines, compresses, or warm-starts from directions WMRL has already identified and thus its update sits largely within Δ_{WMRL} ’s subspace. If true, keeping only the Δ_{WMRL} -aligned component of $\Delta\Delta_{\text{PRL}}$ should recover most of the gain.

Account B (orthogonal / complementary). The policy stage complements WMRL, learning an update in *different* directions of weight space. If true, most of $\Delta\Delta_{\text{PRL}}$ ’s task value should lie in the orthogonal complement of Δ_{WMRL} ’s leading directions.

We test both accounts through left/right subspace projections of $\Delta\Delta_{\text{PRL}}$ and Δ_{PRL} (§5.1–5.2), and module-wise principal-angle overlap (§5.4).

5.1 Left-space (Output-direction) Projection

We compute the SVD $\Delta_{\text{WMRL}} = U\Sigma V^\top$ and form $U_k = U_{:,1:k}$. For each $k \in \{1, 25, 50, 150, 250, 500\}$, we replace $\Delta\Delta_{\text{PRL}}$ with one of three projected variants: projection $U_k U_k^\top \Delta\Delta_{\text{PRL}}$, $(I - U_k U_k^\top) \Delta\Delta_{\text{PRL}}$, $R_k R_k^\top \Delta\Delta_{\text{PRL}}$ (parallel, orthogonal, random control). We then re-evaluate $\theta_0 + \Delta_{\text{WMRL}} + \text{proj}(\Delta\Delta_{\text{PRL}})$.

Finding 1: Policy-stage output directions are largely orthogonal to Δ_{WMRL} . Removing Δ_{WMRL} ’s top- k left-singular directions from $\Delta\Delta_{\text{PRL}}$ has essentially no cost on either split (Fig. 2, red hollow series): at $k=500$, ID 91.7 and OOD 84.9, both at or slightly above the un-projected WMRL $\rightarrow$ PRL baseline (ID 87.0, OOD 81.8). The bulk of $\Delta\Delta_{\text{PRL}}$ ’s task-relevant signal therefore lives outside Δ_{WMRL} ’s principal *output* subspace.

Finding 2: Δ_{WMRL} ’s U directions carry only marginal policy-relevant signal. Keeping only

Δ_{WMRL} ’s top- k left-singular directions in $\Delta\Delta_{\text{PRL}}$ (red dashed series) recovers performance gradually, reaching ID 60.4 / OOD 56.8 at $k=500$. This trails the orthogonal projection by ~ 30 points on *both* splits; a matched random projection (gray) reaches ID 50.3 / OOD 49.5, only ~ 8 points lower. The output directions WMRL has populated are only mildly enriched in policy-relevant information.

Finding 3: U -disjointness is likely intrinsic to policy training, not only two-stage. The pattern holds when we project Δ_{PRL} (PRL trained directly from base, never conditioned on Δ_{WMRL} ; blue series). Orthogonal projection preserves performance (at $k=500$, ID 84.9 / OOD 77.6, vs. un-projected PRL at ID 84.6 / OOD 71.9 – in fact slightly *above* PRL on OOD). Parallel projection recovers only ID 32.8 / OOD 32.3 at $k=1$ and ID 48.2 / OOD 55.2 at $k=500$. Because Δ_{PRL} is computed independently of Δ_{WMRL} , the orthogonality cannot be a consequence of sequential training; it likely reflects an intrinsic difference between the preferred *output* subspaces of policy and world-model learning. Per-checkpoint numbers underlying Fig. 2 are reported in Appendix A.4 (Tabs. 6 and 7).

5.2 Right-space (Input-feature) Projection

§5.1 only examines *output* alignment. Two updates can be orthogonal as matrices and still touch the same input features, if the rotation between their right bases happens to land on different columns of U . We therefore repeat the analysis on the right side, projecting onto Δ_{WMRL} ’s leading *right* singular vectors $\Delta\Delta_{\text{PRL}} V_k V_k^\top$ and $\Delta\Delta_{\text{PRL}} (I - V_k V_k^\top)$.

Finding 4: In contrast, the two stages *share* the right/ V (input-feature) subspace. *Parallel* projection on the right (Fig. 3) is dramatically more effective than on the left: $\Delta\Delta_{\text{PRL}}$ projected onto Δ_{WMRL} ’s top right subspace gives ID 44.0 / OOD 40.6 at $k=1$ and climbs to ID 83.6 / OOD 79.7 at $k=500$ – within a few points of the un-projected WMRL $\rightarrow$ PRL baseline (ID 87.0 / OOD 81.8) on both splits. The same operation on the left only reached ID 60.4 / OOD 56.8 at $k=500$. The random control stays near the WMRL baseline, ruling out a generic rank- k effect. This suggests $V_{\Delta_{\text{WMRL}}}$ **encodes task-relevant information.**

Finding 5: Removing V from Δ_{PRL} hurts performance, but not for $\Delta\Delta_{\text{PRL}}$, suggesting $\Delta\Delta_{\text{PRL}}$ learns robust V -redundant pathways. For $\Delta\Delta_{\text{PRL}}$, removing Δ_{WMRL} ’s leading right singu-

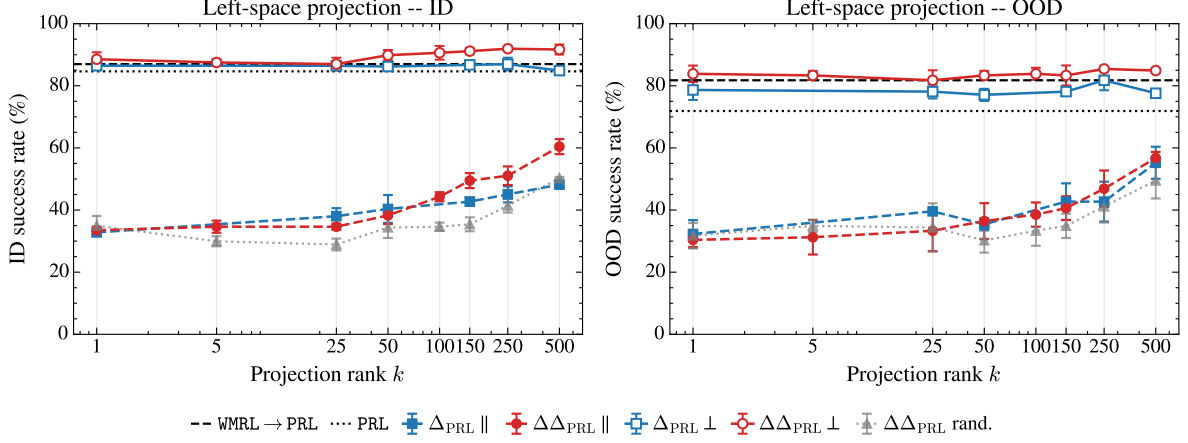


Figure 2: **Left-space (output-direction) projection on ALFWorld, reported separately on ID and OOD.** The two panels share a legend and overlay both projector variants on each policy update: parallel (dashed, filled marker) keeps the Δ_{WMRL} -aligned component; orthogonal (solid, hollow marker) keeps its complement. Blue series correspond to Δ_{PRL} , red to $\Delta\Delta_{\text{PRL}}$; gray is the random-subspace control for $\Delta\Delta_{\text{PRL}}$. Black dashed and dotted lines denote $\text{WMRL} \rightarrow \text{PRL}$ and PRL baselines, respectively, with values taken from the same ID or OOD split as the panel.

lar directions barely hurts: ID $91.4 \rightarrow 89.6$ (nearly flat) and OOD $82.8 \rightarrow 77.6$ (mild drop) from $k=1$ to $k=500$. Taken with Finding 4, this suggests that $\Delta\Delta_{\text{PRL}}$'s task value is redundant across Δ_{WMRL} 's leading right directions and their complement rather than confined to either: removing one side appears to leave a functioning input pathway on the other. However, on Δ_{PRL} , removing the same directions makes performance drop from ID 87.0 / OOD 79.7 at $k=1$ to ID 65.9 / OOD 66.2 at $k=500$, well below the un-projected PRL baseline (ID 84.6 / OOD 71.9) on both splits. Because Δ_{PRL} is trained directly from base with no exposure to Δ_{WMRL} , we infer that PRL -from-base independently discovers a subspace that overlaps with Δ_{WMRL} 's, to the point that projecting Δ_{WMRL} 's right directions out of Δ_{PRL} removes content the policy update needs to function. This adds evidence for Account A on the right side. Per-checkpoint numbers underlying Fig. 3 are in Appendix A.4 (Tabs. 8 and 9).

5.3 A Unified View of Two-stage Agentic Post-Training Geometry

Together, these results point toward a coherent picture of how the two stages interact: Δ_{WMRL} concentrates its update in a low-rank right subspace V that also contains the policy stage's task-relevant input signal (Finding 4), policy training (even independently) writes outputs in U -directions disjoint from WMRL (Findings 1–3), and **sequential training composes the two components without geometric interference** (Finding 5). In other words, the

geometrically identifiable value of sequential training is primarily V -redundancy in the downstream policy Δ . Accounts A and B thus hold *in tandem* on different sides. Though this establishes their weight-space roles, it does not explain how these differences manifest at deployment, or what complementarity accomplishes behaviorally. We next seek behavioral evidence to fill out this picture.

5.4 Module-wise Subspace Overlap

We show in Fig. 4 that the global picture also holds at the module level. For every pair of corresponding weight matrices in Δ_{PRL} and $\Delta\Delta_{\text{PRL}}$, we compute the principal-angle overlap $\bar{\sigma}^2 = \frac{1}{k} \sum_i \sigma_i^2(B_A^\top B_B)$ between their leading- k left (or right) subspaces. The top panel shows the left-side overlap ($k=1$, mean 0.04), nearly orthogonal, while the bottom panel shows the right-side overlap ($k=1$, mean 0.63), highly aligned at dominant input direction.

5.5 Cross-Benchmark Replication

On τ^2 -Bench (with Qwen3-8B), the U -disjoint and V -shared geometric pattern replicates. Rank-1 left-subspace overlap remains near-zero, whereas the right-subspace overlap appears even more strongly than on ALFWorld. However, this is associated more modestly with end-to-end projection performance. The right-parallel projection recovers 71.8% of oracle improvement on ALFWorld vs. 46.9% on τ^2 -Bench, and we thus interpret the geometric replication as stronger than the behavioral evidence (full results in Appendix B).

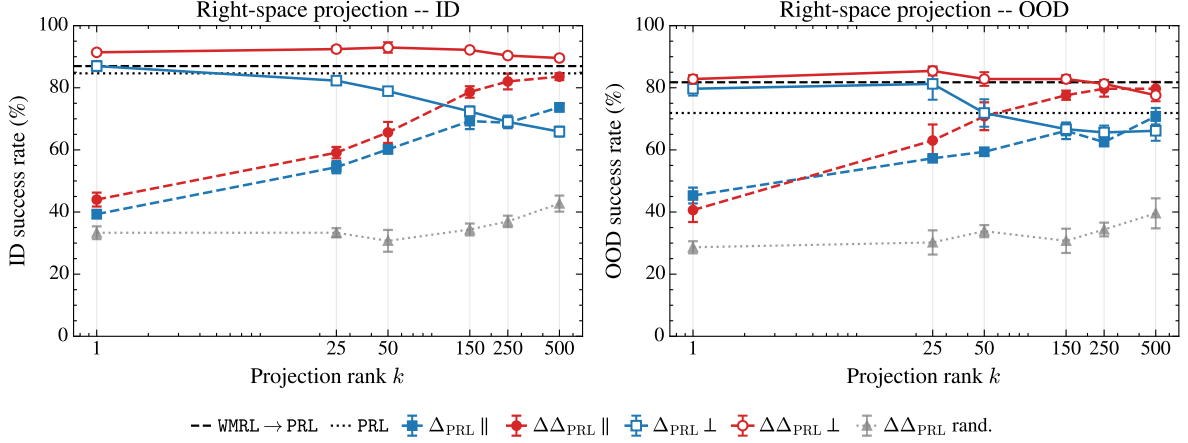


Figure 3: **Right-space (input-feature) projection on ALFWorld, reported separately on ID and OOD.** Same plotting conventions as Fig. 2: the two panels share a legend; parallel projection is shown with dashed filled markers, and orthogonal projection with solid hollow markers. Blue series correspond to Δ_{PRL} , red to $\Delta\Delta_{\text{PRL}}$, and gray to the random-subspace control. Black dashed and dotted lines denote $\text{WMRL} \rightarrow \text{PRL}$ and PRL baselines, respectively. The picture is mirrored relative to the left: orthogonal removal hurts Δ_{PRL} monotonically with k , while parallel projection of $\Delta\Delta_{\text{PRL}}$ recovers most of the $\text{WMRL} \rightarrow \text{PRL}$ baseline by $k=500$ on both splits.

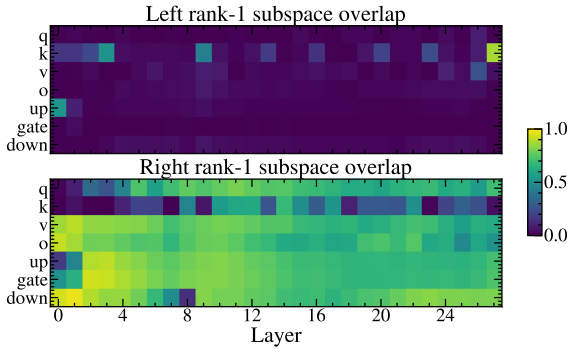


Figure 4: **Module-wise rank-1 subspace overlap** between Δ_{PRL} and $\Delta\Delta_{\text{PRL}}$ on the left (output) vs. right (input-feature) side.

6 How does WMRL help in practice? A rollout-time behavioral account (RQ1)

The geometric findings raise a behavioral question: what does the WMRL-derived structure let the agent do that PRL alone cannot? We probe this through success-set overlap and rollout-time statistics.

Success-set overlap. For each pair of checkpoints we compare tasks solved at pass@1 (Fig. 5). The composite $\text{WMRL} \rightarrow \text{PRL}$ model does *not* subsume either WMRL’s or PRL’s success set: each solves a non-trivial number of tasks WMRL $\rightarrow$ PRL fails on, while $\text{WMRL} \rightarrow \text{PRL}$ itself picks up new successes that neither base achieves. This non-nesting rules out the simplest reading, i.e. that $\text{WMRL} \rightarrow \text{PRL}$ is simply a refined version of PRL, since refinement would predict success-set containment. This sug-

Pairwise success overlap @ pass@1 ($N=192$)				
Base	27			
WMRL	18	64		
PRL	23	58	154	
WMRL $\rightarrow$ PRL	27	61	139	163
	Base	WMRL	PRL	WMRL $\rightarrow$ PRL

Figure 5: Success-set overlap at pass@1 on ALFWorld. Diagonal = #samples a model passes; off-diagonal = #samples both pass.

gests that what the second stage is doing on top of WMRL is not a near-copy of what PRL does on its own and instead points to synergistic effects.

Exploration statistics. Table 2 reports per-task averages of unique valid actions, unique observation states, and total turns across rollouts, with each run capped at 30 turns. Uniqueness is computed by exact string match, and an action is valid only if it changes the environment state. We also report the three-run union as a cross-check.

WMRL does not improve unique-action coverage over the base model, but maintains broader *state* exploration. PRL, in contrast, sharply contracts both action and state coverage in exchange for shorter, more focused trajectories. The composite $\text{WMRL} \rightarrow \text{PRL}$ inherits PRL’s efficiency (similar turn count) while recovering some of WMRL’s state-coverage advantage, suggesting that the exploration benefit of WMRL is preserved *behaviorally* through

Table 2: Exploration statistics on ALFWorld (ID+OOD, $N=192$). We report per-task averages of unique valid actions, unique states, and total turns.

Model	Pass@1			Pass@3 Union		
	N_{act}	N_{state}	Turns	N_{act}	N_{state}	Turns
Base	10.8	14.8	28.0	18.3	26.7	82.0
WMRL	7.8	13.5	24.9	13.3	26.5	74.6
PRL	7.1	8.1	12.6	8.8	10.3	37.7
WMRL $\rightarrow$ PRL	8.4	9.7	12.4	11.6	13.9	37.5

the policy stage. This rules out WMRL $\rightarrow$ PRL being simply a more efficient WMRL or a more reward-optimized PRL since each of the baselines lacks one of the two behavioral axes the composite has.

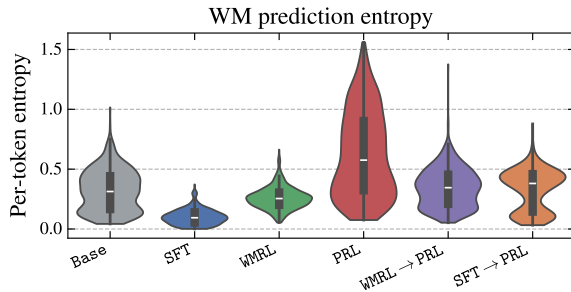


Figure 6: Per-token entropy of each checkpoint’s distribution over the *ground-truth* next-state tokens, evaluated on the WMRL held-out test set ($N=512$ single-turn transitions) and averaged per sample.

Next-state prediction entropy on held-out WM set. Fig. 6 reports per-token entropy on a held-out WMRL evaluation set, scored against the *ground-truth* next-state tokens. The prompt is slightly revised to directly predict the next state without thinking for fair comparison, same as WMSFT. Lower indicates the models are better calibrated as a world model. We observe (i) WMRL reduces next-state entropy relative to base, i.e. WMRL training makes the model a more calibrated next-state predictor, consistent with its training objective (ii) PRL-from-base *degrades* next-state calibration, by itself actively pushing the model *away* from accurate next-state prediction, and (iii) WMRL $\rightarrow$ PRL preserves most of WMRL’s improvement. Here, the second-stage policy update does *not* erase the calibration that the first-stage WMRL update put in place.

Summary Taken together, the three behavioral findings point in the same direction: WMRL $\rightarrow$ PRL is an effective combination of two qualitatively different stages. The non-nested success sets show

this at the *task* level (each baseline solves tasks the composite misses, and vice versa). The exploration statistics show it at the *trajectory* level (the composite combines WMRL’s state coverage with PRL’s efficiency). Finally, the entropy results show it at the *representational* level (the composite retains the WM calibration that PRL-from-base implicitly degrades). We hypothesize that these three patterns are *plausible* (observational) behavioral correlates of the §5 geometry, in which the two stages write to disjoint output subspaces (Findings 1–3) while sharing input subspaces (Finding 4), although we cannot directly establish a causal relationship. This connection is most direct for entropy preservation, since orthogonal weight updates do not overwrite each other’s columns by construction. Though these properties are not uniquely predicted by the geometric patterns, they point toward future causal investigations to better characterize such effects.

7 Preserving the World Model (RQ3)

The asymmetric geometry of §5 predicts that gentle preservation of Δ_{WMRL} during policy training could help performance, and that extra world-model signal should compose constructively with the policy update. We test this assumption with two interventions, one training-free and one training-time (for full details, refer to Appendix C):

Training-free: KnOTS / TIES merging of M_{WMRL} and the policy model. We additively merge M_{WMRL} with either M_{PRL} (PRL from base) or $M_{\text{WMRL}+\text{PRL}}$ (sequential WMRL $\rightarrow$ PRL) via TIES (Yadav et al., 2023) and KnOTS (Stoica et al., 2025). KnOTS factors out a singular basis shared across tasks before applying the TIES mask; since our geometry has V shared and U disjoint, we test both the default U -sharing and a V -sharing variant.

Training-based: online WM-SFT during PRL. During the second stage we additionally subsample 32 realized transitions from each step’s rollouts to compute a next-state SFT loss on top of the policy gradient, mirroring the ECHO recipe of concurrent work by Shrivastava et al. (2026), which learns a world model “for free” inside terminal-agent RL.

Interventions improve over the untreated WMRL $\rightarrow$ PRL baseline. On $(M_{\text{WMRL}}, M_{\text{PRL}})$, KnOTS- V beats KnOTS- U by +2.0 (+4.2 OOD), matching the V -shared/ U -disjoint prediction: factoring out the basis that the two tasks actually share (V) preserves signal that the U basis variant loses.

Table 3: Preservation interventions on ALFWorld. For each base pair (PRL or WMRL $\rightarrow$ PRL as the second task) we list the source checkpoints, plain TIES merging, and KnOTS-TIES in both the default U -sharing and our V -sharing variant. The bottom row is the training-based on-line WMSFT auxiliary loss during PRL. **Bold/underlined** indicates the best/baseline results within each sub-block.

Method	ID	OOD	AVG
base	19.8 ± 4.5	15.6 ± 1.3	17.7 ± 2.9
WMRL	33.6 ± 3.4	31.8 ± 5.3	32.7 ± 3.3
PRL	<u>84.6 ± 1.3</u>	<u>71.9 ± 2.6</u>	<u>78.3 ± 0.9</u>
<i>training free</i>			
TIES	85.0 ± 1.7	72.6 ± 1.2	78.8 ± 0.5
KnOTS (U)	88.0 ± 0.7	74.5 ± 3.2	81.2 ± 1.7
KnOTS(V)	87.8 ± 2.2	78.7 ± 1.9	83.2 ± 1.9
WMRL $\rightarrow$ PRL	<u>87.0 ± 0.4</u>	<u>81.8 ± 0.7</u>	<u>84.4 ± 0.3</u>
<i>training free</i>			
TIES	87.5 ± 1.3	82.0 ± 0.4	84.8 ± 1.0
KnOTS(U)	89.8 ± 1.1	82.8 ± 1.3	86.3 ± 0.3
KnOTS(V)	90.1 ± 1.0	81.8 ± 1.9	85.9 ± 0.7
<i>training based</i>			
Online WMSFT	90.4 ± 0.4	79.5 ± 0.7	85.0 ± 0.5

On $(M_{\text{WMRL}}, M_{\text{WMRL}+\text{PRL}})$, the V/U gap flips but is very close; because $\Delta_{\text{WMRL}+\text{PRL}} = \Delta_{\text{WMRL}} + \Delta\Delta_{\text{PRL}}$ already incorporates Δ_{WMRL} , reducing the effect of the basis choice. Both merge variants still sit ~ 2 points above the untreated baseline on ALFWorld. Finally, online WMSFT *during* PRL reaches the highest ID (90.4) but trails on OOD (79.5). The two interventions therefore complement each other: a training-free merger built on the right geometric basis (V) recovers most of the gain without any further optimization, and a light training-time WMSFT signal further improves ID performance.

8 Conclusion

The composition of world models and policies is moving from an emergent property of large models to an explicit training criterion. This shift opens two questions. One is *engineering*: how should the stages be combined to produce capable agents? The other is scientific: how does a single network meaningfully compose both kinds of knowledge, and what structural properties does this produce? We offer an initial pathway for investigating both. In our experiments, we find that world-model and policy training share input-feature directions and write to largely distinct output directions, and also that sequential training makes the resulting policy less dependent on the world model’s leading input directions. These findings suggest ways forward

for both investigating and engineering how these two stages compose.

Limitations. Our analysis is empirical and confined to two text-only agent benchmarks and two Qwen2.5/Qwen3 models. This setting is less general than more realistic web- or computer-use agents, which often require vision-language perception, grounding, and interaction. We thus view our work as a first exploration of these phenomena. However, world-model training and evaluation for multimodal agents remain comparatively immature, making text-only environments a useful controlled setting for isolating the geometry of world-model preservation. The left/right asymmetry we identify is robust across modules and depths within these settings, but its mechanistic origin remains open. One possible contributor is a difference between target distributions of world-model and policy training. However, this pattern also appears on τ^2 -Bench, where both of these stages contain natural-language dialogue and tool-related content. Still, our experiments cannot yet isolate the contribution of target-distribution differences.

We also restrict our analysis to additive parameter updates relative to a single base model; extending the framework to LoRA-style decomposed updates (Hu et al., 2022) is a natural next step. Meanwhile, our interventions are module-wise but apply the same intervention to all modules. Stronger causal tests, such as cross-module analysis (Wang et al., 2023), layer-local editing (Geva et al., 2021), activation patching (Meng et al., 2022), or training-time constraints on the U/V subspaces (Zou et al., 2025), are left for future work.

Finally, our preservation experiments instantiate only a small part of the design space suggested by the geometry. We evaluate KnOTS/TIES merging and an online WM-SFT auxiliary loss, but do not extensively tune merge hyperparameters; on τ^2 -Bench, we replicate only the training-free merge experiments due to the cost of running on-line WMSFT. More scalable preservation methods and broader benchmark/model coverage remain important next steps. We hope this exploratory study motivates more systematic investigations of world-model preservation through geometry, especially in multimodal agents, and inspires scalable methods for boosting agentic post-training.

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A ALFWorld: Implementation Details and Extended Results

This appendix collects (i) implementation details and hyperparameters for the ALFWorld experiments, (ii) the WMRL configuration sweep that produced the cluster of checkpoints in Fig. 1, (iii) the module-level effective-rank breakdown, (iv) magnitude-pruning controls referenced in §A.5, and (v) the full per-checkpoint performance tables behind the projection and pruning figures in the main text. Throughout we follow the experimental setup of Yu et al. (2026) as closely as possible: all checkpoints are trained from publicly released base models with GRPO for both the WMRL and PRL stages.

A.1 Training and Reproduction Setup

Base model and data. We start from Qwen2.5-7B-Instruct (Qwen Team, 2025a). The WMRL training corpus is collected by rolling out the base model on the official ALFWorld training split ($N=3$ trajectories per task, temperature 1.0, max 30 steps), converting trajectories into per-step transition triplets $\langle s_{t-H:t}, a_t, s_{t+1} \rangle$, and discarding samples with invalid actions. Following Yu et al. (2026), we fix $H = 2$ for training efficiency and perform SFT with a held-out split to get a filter model, which we then use to subsample “too-easy” triplets retaining $\sim 30\%$ of the original. The data size is $\sim 16K$.

Sim-to-real reward. We use Qwen3-Embedding-8B (Zhang et al., 2025) as the frozen encoder $E(\cdot)$ with cosine-similarity threshold $\eta = 0.8$.

WMRL stage. 2 epochs, learning rate $1e-6$, effective batch size 32, GRPO group size 8. Each checkpoint is trained on two NVIDIA H200 GPUs for around 8 hours.

PRL stage. GRPO with terminal task-success reward, discount $\gamma = 1.0$, maximum 15 steps per episode during training, 300 update steps, group size 8. We use the official repo of VerI-Agent (Feng

et al., 2025) for the multi-turn training¹. Each checkpoint is trained on two NVIDIA H200 GPUs for around 30 hours.

A.2 WMRL Setting Sweep

To probe how sensitive WMRL is to its training-time choices—and to assemble the cluster of WMRL configurations plotted in Fig. 1—we sweep along four axes with anchor values ($\eta = 0.8$, self-rollouts, single-turn prediction, binarized embedding score):

- **Reward threshold** $\eta \in \{0.5, 0.8, 0.9\}$, controlling how strict the similarity criterion is.
- **Data source:** self-rollouts vs. expert demonstrations from ALFWorld official data.
- **Prediction format:** single-turn vs. three consecutive steps’ predictions with averaged reward (multi-state predictions with multiple corresponding actions provided)
- **Judge category:** LLM-as-judge (Zheng et al., 2023) with Qwen3-235B-A22B-Instruct vs. Qwen3-8B embedding score.

Table 4: WMRL setting sweep on ALFWorld (% , AVG). Each row reports the standalone WMRL checkpoint and the downstream WMRL $\rightarrow$ PRL pipeline.

Setting	WMRL	WMRL $\rightarrow$ PRL
<i>Anchor</i>		
-	32.7 $\pm$ 3.3	84.4 $\pm$ 0.3
<i>Reward threshold</i>		
$\eta = 0.5$	10.2 $\pm$ 3.4	71.9 $\pm$ 2.0
$\eta = 0.9$	18.2 $\pm$ 2.6	76.1 $\pm$ 0.8
<i>Data source</i>		
Expert demonstrations	16.4 $\pm$ 3.4	74.0 $\pm$ 1.3
<i>Prediction format</i>		
Multi-turn	21.5 $\pm$ 1.2	75.6 $\pm$ 3.1
<i>Judge Category</i>		
LLM-as-a-Judge	17.2 $\pm$ 2.8	78.7 $\pm$ 1.9

A.3 Per-layer Spectral Picture of the Pipeline

At every layer, the post-training updates show a clear spectral separation: Δ_{WMRL} above $\Delta_{\text{WMRL}+\text{PRL}}$ above Δ_{PRL} , with Δ_{WMSFT} far below all three (Fig. 7). We use this layer-wise ordering primarily for descriptive context since many plausible accounts (e.g., shared gradient dynamics across modules) would predict module-wise consistency.

¹<https://github.com/langfengq/ver1-agent>

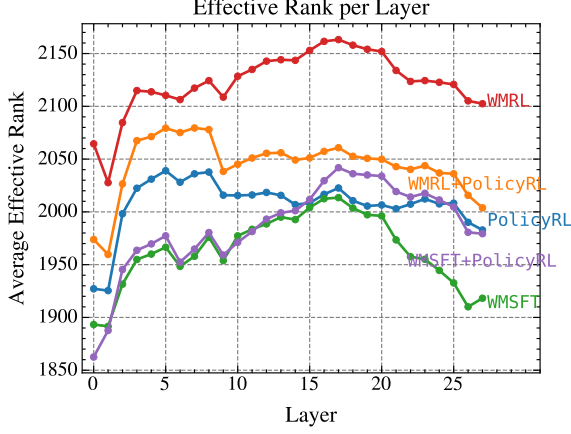


Figure 7: Per-layer effective rank of the parameter updates produced by each post-training paradigm, averaged across modules. The ordering of paradigms is preserved across depth.

A.4 Full Performance Tables for the Projection

The main text reports ID and OOD success rates separately for the projection (§5) and pruning (§A.5, Appendix A.6) experiments. For completeness, this appendix collects the full ID, OOD, and AVG (= Unweighted mean over ID and OOD) numbers for every checkpoint used in those analyses, shown as the mean $\pm$ standard deviation over 3 roll-out repeats. Baselines from Tab. 1 are reproduced in Tab. 5 for convenience.

Table 5: ALFWorld baseline reference checkpoints (reproduced from Tab. 1 for cross-reference).

ckpt	ID	OOD	AVG
ReAct(base)	19.8 $\pm$ 4.5	15.6 $\pm$ 1.3	17.7 $\pm$ 2.9
WMSFT	5.0 $\pm$ 1.0	2.1 $\pm$ 1.5	3.5 $\pm$ 1.2
WMRL	33.6 $\pm$ 3.4	31.8 $\pm$ 5.3	32.7 $\pm$ 3.3
PRL	84.6 $\pm$ 1.3	71.9 $\pm$ 2.6	78.3 $\pm$ 0.9
WMSFT+PRL	66.4 $\pm$ 1.3	55.2 $\pm$ 2.0	60.8 $\pm$ 1.5
WMRL+PRL	87.0 $\pm$ 0.4	81.8 $\pm$ 0.7	84.4 $\pm$ 0.3

Left-space projection (§5.1). Tab. 6 reports Δ_{PRL} projected onto Δ_{WMRL} ’s top- k left singular subspace; Tab. 7 reports $\Delta\Delta_{\text{PRL}}$ projected onto the same.

Right-space projection (§5.2). Tab. 8 reports Δ_{PRL} projected onto Δ_{WMRL} ’s top- k right singular subspace; Tab. 9 reports the same for $\Delta\Delta_{\text{PRL}}$.

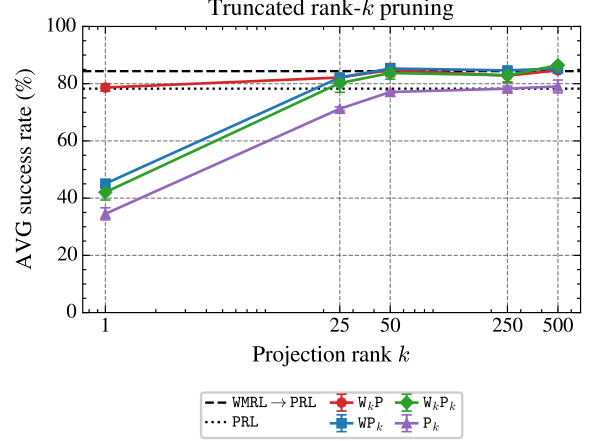


Figure 8: **Rank- k truncation pruning of Δ s.** Across $k \in \{1, 25, 50, 250, 500\}$, pruning Δ_{WMRL} is nearly harmless, whereas pruning $\Delta\Delta_{\text{PRL}}$ or Δ_{PRL} causes a sharp small-rank drop.

A.5 Robustness Check: rank- k Pruning Experiments

The geometric picture of §5 – Δ_{WMRL} suggests a low-rank, input-aligned scaffold that the policy stage builds on through output-orthogonal directions – makes a prediction about each stage’s update in isolation: Δ_{WMRL} should be highly truncation-tolerant, while Δ_{PRL} – a from-base update that has to rediscover the same hypothetical scaffold – should not be. We test these predictions through rank- k truncation, following prior work on low-rank approximations of neural network weights for compression and adaptation (Denton et al., 2014; Aghajanyan et al., 2021; Hu et al., 2022) and on rank-based compression of LLM weights (Ashkboos et al., 2024; Sharma et al., 2024).

Concretely, for each k and each $\Delta \in \{\Delta_{\text{WMRL}}, \Delta\Delta_{\text{PRL}}, \Delta_{\text{PRL}}\}$ we form the truncated rank- k approximation Δ_k and evaluate four variants: W_kP , which prunes Δ_{WMRL} only; WP_k , which prunes $\Delta\Delta_{\text{PRL}}$ only; W_kP_k , which prunes both updates; and P_k , which directly prunes the from-base update Δ_{PRL} . Tab. 10 reports the per- k ID, OOD, and AVG numbers for the four rank-truncation regimes and Fig. 8 displays the trend.

Δ_{WMRL} is highly compressible. W_kP is almost flat across ranks (Fig. 8): even replacing Δ_{WMRL} by its rank-1 truncation while keeping $\Delta\Delta_{\text{PRL}}$ intact yields ID 77.6 / OOD 79.7, within ~ 10 points of the full $\text{WMRL} \rightarrow \text{PRL}$ model (ID 87.0, OOD 81.8). Thus, most of the downstream utility of the WM stage is concentrated in very few directions.

Table 6: Left-space projection of Δ_{PRL} onto Δ_{WMRL} . Composition: $\theta_0 + \text{proj}_L(\Delta_{\text{PRL}})$. We report performance for the parallel component ($UU^\top \Delta_{\text{PRL}}$) and the orthogonal component ($(I - UU^\top) \Delta_{\text{PRL}}$).

k	<i>parallel</i>			<i>orthogonal</i>		
	ID	OOD	AVG	ID	OOD	AVG
1	32.8±1.1	32.3±4.5	32.5±2.7	86.5±1.3	78.7±3.2	82.5±2.2
25	38.0±2.6	39.6±0.7	38.8±1.5	86.5±0.7	78.1±2.2	82.3±0.7
50	40.4±4.5	35.4±1.9	37.9±1.7	86.2±0.4	77.1±1.9	81.6±0.8
150	42.7±1.3	42.7±5.9	42.7±3.2	86.7±0.6	78.1±1.3	82.4±1.0
250	45.0±2.6	42.7±6.4	43.9±3.1	87.0±2.0	81.8±3.2	84.4±2.6
500	48.2±1.0	55.2±5.2	51.7±2.7	84.9±0.7	77.6±0.7	81.2±0.6

Table 7: Left-space projection of $\Delta\Delta_{\text{PRL}}$ onto Δ_{WMRL} . Composition: $\theta_0 + \Delta_{\text{WMRL}} + \text{proj}_L(\Delta\Delta_{\text{PRL}})$. Columns report the parallel component ($UU^\top \Delta\Delta_{\text{PRL}}$), orthogonal component ($(I - UU^\top) \Delta\Delta_{\text{PRL}}$), and random projection ($QQ^\top \Delta\Delta_{\text{PRL}}$ with Q random orthonormal).

k	<i>parallel</i>			<i>orthogonal</i>			<i>random</i>		
	ID	OOD	AVG	ID	OOD	AVG	ID	OOD	AVG
1	33.5±1.2	30.4±2.3	32.0±0.8	88.5±2.2	83.8±2.7	86.2±1.0	34.9±3.2	31.8±4.1	33.3±1.5
5	34.6±1.9	31.2±5.6	32.9±3.5	87.5±1.1	83.3±1.5	85.4±1.1	29.9±1.6	34.9±1.9	32.4±1.7
25	34.6±1.0	33.3±6.5	34.0±3.4	87.0±2.0	81.8±3.2	84.4±1.7	28.9±1.9	34.4±7.8	31.6±3.9
50	38.3±2.8	36.5±5.8	37.4±4.2	89.8±1.7	83.3±1.5	86.6±0.7	34.4±3.4	30.2±3.9	32.3±1.0
100	44.3±1.5	38.5±3.9	41.4±2.2	90.6±2.2	83.8±1.9	87.2±2.0	34.6±1.3	33.3±4.8	34.0±2.2
150	49.5±2.4	40.6±3.8	45.0±1.2	91.2±1.0	83.3±3.2	87.2±2.1	35.4±2.2	34.9±3.9	35.2±2.6
250	51.0±3.0	46.9±5.8	49.0±3.3	91.9±0.7	85.4±0.7	88.7±0.6	41.4±2.2	41.1±5.2	41.3±3.7
500	60.4±2.4	56.8±1.9	58.6±1.8	91.7±1.6	84.9±0.7	88.3±1.1	50.3±0.4	49.5±5.8	49.9±2.7

$\Delta\Delta_{\text{PRL}}$ and Δ_{PRL} **require higher rank**. In contrast, WP_k drops to ID 43.8 / OOD 46.4 at $k=1$ and recovers only by moderate ranks, showing that $\Delta\Delta_{\text{PRL}}$ spreads its task value across many singular components. W_kP_k closely tracks WP_k , indicating that $\Delta\Delta_{\text{PRL}}$, rather than Δ_{WMRL} , is the bottleneck. The direct from-base update is densest: P_k is worst at every k , with rank-1 performance near the WMRL-only model and rank 500 only matching unpruned PRL.

Robustness with respect to the geometric picture. These trends match the geometry in §5: Δ_{WMRL} provides a low-rank scaffold that the policy stage can reuse, whereas PRL from base must construct that scaffold and the task-specific correction jointly. The similar pattern appears under magnitude pruning: top- p pruning recovers $\Delta\Delta_{\text{PRL}}$ to the WMRL $\rightarrow$ PRL baseline best with only 10% of entries retained, while pruning Δ_{PRL} yields best performance with 50% (Appendix A.6).

A.6 Magnitude Pruning of Δ s

As a coordinate-space complement to the spectral analyses, we apply classical magnitude pruning (Han et al., 2015; Frankle and Carbin, 2019) – also a central primitive in recent LLM weight pruning methods such as SparseGPT (Frantar and

Alistarh, 2023), Wanda (Sun et al., 2024) and LLM-Pruner (Ma et al., 2023) – to Δ_{PRL} and $\Delta\Delta_{\text{PRL}}$, retaining only a p fraction of the Δ entries (zeroing the rest) chosen by three different selection rules and re-evaluating task success: (i) *top- p* , keeping the entries with the largest absolute magnitude; (ii) *bottom- p* , keeping the entries with the smallest absolute magnitude (i.e. the lowest-magnitude tail); (iii) *random- p* , keeping a random p fraction. The latter two serve as baselines that isolate, respectively, the effect of discarding the high-magnitude head and the effect of total mass alone. Per- p ID, OOD, and AVG numbers for all three selection rules are in Tab. 11. The sign_only row keeps only the sign of every entry and rescales magnitudes to a fixed constant. The curves are also shown in Fig. 9 to better track the trend.

Three patterns are consistent with the spectral story: (i) top- p on $\Delta\Delta_{\text{PRL}}$ reaches the best performance with $p = 0.1$, while the curve for Δ_{PRL} still goes up before $p = 0.5$, indicating that the update of $\Delta\Delta_{\text{PRL}}$ is more compact. This is consistent with $\Delta\Delta_{\text{PRL}}$ ’s smaller, output-orthogonal spectral role on top of Δ_{WMRL} . (ii) Bottom- p collapses both Δ_{PRL} and $\Delta\Delta_{\text{PRL}}$ sharply for every p – the high-magnitude entries discarded by bottom- p are precisely the ones that carry the task signal,

Table 8: Right-space projection of Δ_{PRL} onto Δ_{WMRL} . Composition: $\theta_0 + \text{proj}_R(\Delta_{\text{PRL}})$. Columns report the parallel component ($\Delta_{\text{PRL}}VV^\top$), orthogonal component ($\Delta_{\text{PRL}}(I - VV^\top)$), and random projection ($\Delta_{\text{PRL}}QQ^\top$ with Q random orthonormal).

k	<i>parallel</i>			<i>orthogonal</i>			<i>random</i>		
	ID	OOD	AVG	ID	OOD	AVG	ID	OOD	AVG
1	39.3 $\pm$ 1.3	45.3 $\pm$ 2.5	42.3 $\pm$ 1.8	87.0 $\pm$ 1.6	79.7 $\pm$ 2.2	83.3 $\pm$ 1.8	31.8 $\pm$ 0.4	28.6 $\pm$ 1.9	30.2 $\pm$ 1.0
25	54.4 $\pm$ 1.9	57.3 $\pm$ 0.7	55.9 $\pm$ 0.6	82.3 $\pm$ 0.7	81.2 $\pm$ 5.1	81.8 $\pm$ 2.6	32.5 $\pm$ 1.6	32.3 $\pm$ 1.5	32.4 $\pm$ 0.3
50	60.2 $\pm$ 1.3	59.4 $\pm$ 1.3	59.8 $\pm$ 1.1	78.9 $\pm$ 1.1	71.9 $\pm$ 4.4	75.4 $\pm$ 2.0	29.4 $\pm$ 5.1	29.2 $\pm$ 2.7	29.3 $\pm$ 2.5
150	69.3 $\pm$ 2.6	66.2 $\pm$ 2.7	67.7 $\pm$ 0.8	72.4 $\pm$ 1.6	66.7 $\pm$ 1.9	69.5 $\pm$ 1.0	34.1 $\pm$ 1.3	32.8 $\pm$ 1.3	33.5 $\pm$ 1.2
250	68.8 $\pm$ 1.3	62.5 $\pm$ 0.0	65.6 $\pm$ 0.6	69.0 $\pm$ 2.0	65.6 $\pm$ 2.2	67.3 $\pm$ 1.9	37.5 $\pm$ 1.3	35.9 $\pm$ 0.0	36.7 $\pm$ 0.6
500	73.7 $\pm$ 1.3	70.8 $\pm$ 2.7	72.3 $\pm$ 2.0	65.9 $\pm$ 1.6	66.2 $\pm$ 3.2	66.0 $\pm$ 1.4	41.7 $\pm$ 2.6	37.5 $\pm$ 2.2	39.6 $\pm$ 1.8

Table 9: Right-space projection of $\Delta\Delta_{\text{PRL}}$ onto Δ_{WMRL} . Composition: $\theta_0 + \Delta_{\text{WMRL}} + \text{proj}_R(\Delta\Delta_{\text{PRL}})$.

k	<i>parallel</i>			<i>orthogonal</i>			<i>random</i>		
	ID	OOD	AVG	ID	OOD	AVG	ID	OOD	AVG
1	44.0 $\pm$ 2.2	40.6 $\pm$ 3.8	42.3 $\pm$ 1.6	91.4 $\pm$ 0.6	82.8 $\pm$ 1.3	87.1 $\pm$ 0.8	33.3 $\pm$ 2.0	28.6 $\pm$ 1.9	31.0 $\pm$ 1.8
25	59.1 $\pm$ 1.8	63.0 $\pm$ 5.2	61.1 $\pm$ 2.3	92.5 $\pm$ 1.0	85.4 $\pm$ 1.5	88.9 $\pm$ 1.2	33.3 $\pm$ 1.5	30.2 $\pm$ 3.9	31.8 $\pm$ 1.5
50	65.6 $\pm$ 3.4	70.8 $\pm$ 4.5	68.2 $\pm$ 1.3	93.0 $\pm$ 1.7	82.8 $\pm$ 2.2	87.9 $\pm$ 0.3	30.7 $\pm$ 3.5	33.9 $\pm$ 1.9	32.3 $\pm$ 1.9
150	78.7 $\pm$ 1.8	77.6 $\pm$ 1.5	78.1 $\pm$ 1.7	92.2 $\pm$ 0.0	82.8 $\pm$ 1.3	87.5 $\pm$ 0.6	34.4 $\pm$ 1.9	30.7 $\pm$ 3.9	32.5 $\pm$ 1.0
250	82.0 $\pm$ 2.5	79.7 $\pm$ 2.5	80.9 $\pm$ 1.3	90.4 $\pm$ 0.7	81.2 $\pm$ 1.3	85.8 $\pm$ 0.4	37.0 $\pm$ 1.8	34.4 $\pm$ 2.2	35.7 $\pm$ 1.8
500	83.6 $\pm$ 1.1	79.7 $\pm$ 2.2	81.6 $\pm$ 1.5	89.6 $\pm$ 0.7	77.6 $\pm$ 1.9	83.6 $\pm$ 0.6	42.7 $\pm$ 2.6	39.6 $\pm$ 4.8	41.1 $\pm$ 1.4

Table 10: Truncated rank- k pruning of Δ s (§A.5). W_kP : $\theta_0 + (\Delta_{\text{WMRL}})_k + \Delta\Delta_{\text{PRL}}$; WP_k : $\theta_0 + \Delta_{\text{WMRL}} + (\Delta\Delta_{\text{PRL}})_k$; W_kP_k : $\theta_0 + (\Delta_{\text{WMRL}})_k + (\Delta\Delta_{\text{PRL}})_k$; P_k : $\theta_0 + (\Delta_{\text{PRL}})_k$.

Method	k	ID	OOD	AVG
W_kP	1	77.6 $\pm$ 1.0	79.7 $\pm$ 2.2	78.7 $\pm$ 1.3
	25	82.0 $\pm$ 0.6	82.3 $\pm$ 1.9	82.2 $\pm$ 1.3
	50	85.7 $\pm$ 1.6	83.8 $\pm$ 3.0	84.8 $\pm$ 2.2
	250	86.5 $\pm$ 1.8	79.2 $\pm$ 2.7	82.8 $\pm$ 2.2
	500	86.5 $\pm$ 1.0	82.8 $\pm$ 2.2	84.6 $\pm$ 0.8
WP_k	1	43.8 $\pm$ 2.9	46.4 $\pm$ 0.7	45.0 $\pm$ 1.4
	25	85.7 $\pm$ 2.2	78.7 $\pm$ 1.5	82.2 $\pm$ 0.7
	50	88.3 $\pm$ 1.1	82.3 $\pm$ 0.7	85.3 $\pm$ 0.9
	250	88.0 $\pm$ 1.0	81.2 $\pm$ 2.2	84.6 $\pm$ 0.8
	500	88.3 $\pm$ 1.9	82.3 $\pm$ 1.5	85.3 $\pm$ 1.6
W_kP_k	1	40.4 $\pm$ 2.4	43.8 $\pm$ 3.4	42.1 $\pm$ 2.7
	25	81.2 $\pm$ 3.2	79.2 $\pm$ 3.9	80.2 $\pm$ 3.2
	50	85.7 $\pm$ 2.9	81.8 $\pm$ 1.5	83.7 $\pm$ 2.1
	250	85.2 $\pm$ 1.9	80.7 $\pm$ 3.2	82.9 $\pm$ 2.6
	500	88.0 $\pm$ 1.0	84.9 $\pm$ 1.9	86.5 $\pm$ 0.5
P_k	1	34.1 $\pm$ 5.0	34.9 $\pm$ 0.7	34.5 $\pm$ 2.1
	25	77.3 $\pm$ 1.7	65.1 $\pm$ 1.5	71.2 $\pm$ 0.7
	50	81.8 $\pm$ 3.1	72.4 $\pm$ 4.1	77.1 $\pm$ 0.5
	250	84.6 $\pm$ 2.2	71.9 $\pm$ 1.3	78.3 $\pm$ 0.9
	500	84.1 $\pm$ 1.5	74.0 $\pm$ 5.8	79.0 $\pm$ 2.2

complementing the top- p result. (iii) Random- p recovers performance roughly in proportion to mass: $\Delta\Delta_{\text{PRL}}$ random- p matches top- p only as $p \rightarrow 0.5$, while Δ_{PRL} random- p trails top- p by a wide margin at small p . So the gain of top- p over random- p , at any given p , gives a coordinate-level analogue of the spectral concentration story – $\Delta\Delta_{\text{PRL}}$ ’s top-

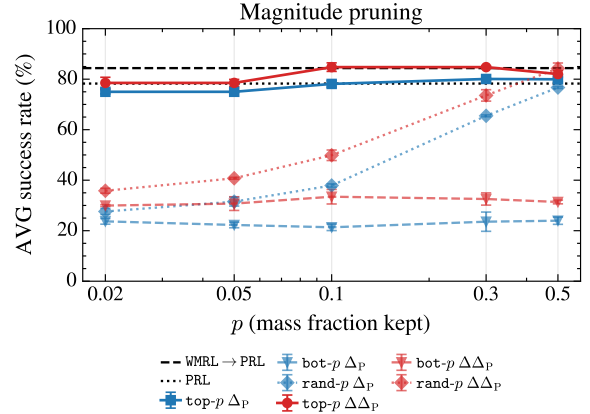


Figure 9: Magnitude pruning on Δ_{PRL} (blue) and $\Delta\Delta_{\text{PRL}}$ (red), comparing the top- p selection rule (solid) against bottom- p (dashed) and random- p (dotted) baselines. Top- p keeps the entries of Δ with the largest absolute magnitude; bottom- p keeps the smallest; random- p is a random subset of the same mass. Top- p on $\Delta\Delta_{\text{PRL}}$ performs best at $p = 0.1$; top- p on Δ_{PRL} needs $p \geq 0.3$ to reach the maxima; bottom- p is near-baseline for all p ; random- p saturates only as $p \rightarrow 0.5$.

magnitude entries dominate downstream performance much more than Δ_{PRL} ’s.

B τ^2 -Bench: Cross-task Replication

This appendix reports a partial replication of the ALFWORLD findings on τ^2 -Bench with a different base model (Qwen3-8B). §B.1 describes the training setup and end-to-end performance numbers,

Table 11: Magnitude pruning of Δ s on ALFWorld. p is the fraction of entries retained. Left block prunes Δ_{PRL} ; right block prunes $\Delta\Delta_{\text{PRL}}$. Each section corresponds to one selection rule, followed by the sign-only baseline.

p	Δ_{PRL} pruned			$\Delta\Delta_{\text{PRL}}$ pruned		
	ID	OOD	AVG	ID	OOD	AVG
<i>top-p</i>						
0.02	75.5 $\pm$ 0.4	74.5 $\pm$ 1.5	75.0 $\pm$ 0.8	82.0 $\pm$ 5.7	75.0 $\pm$ 2.2	78.5 $\pm$ 2.2
0.05	74.5 $\pm$ 1.3	75.5 $\pm$ 0.7	75.0 $\pm$ 0.3	82.0 $\pm$ 3.5	75.0 $\pm$ 2.5	78.5 $\pm$ 1.4
0.10	82.8 $\pm$ 2.2	73.4 $\pm$ 3.4	78.1 $\pm$ 1.3	88.3 $\pm$ 1.3	81.2 $\pm$ 2.2	84.8 $\pm$ 1.7
0.30	85.7 $\pm$ 1.0	74.5 $\pm$ 0.7	80.1 $\pm$ 0.3	88.8 $\pm$ 0.4	80.7 $\pm$ 1.5	84.8 $\pm$ 0.6
0.50	85.4 $\pm$ 1.9	74.5 $\pm$ 0.7	80.0 $\pm$ 1.0	87.0 $\pm$ 1.8	77.1 $\pm$ 3.9	82.0 $\pm$ 1.4
<i>bottom-p</i>						
0.02	24.0 $\pm$ 1.0	23.4 $\pm$ 1.3	23.7 $\pm$ 1.1	32.3 $\pm$ 3.2	27.6 $\pm$ 3.2	29.9 $\pm$ 0.4
0.05	23.7 $\pm$ 0.7	20.8 $\pm$ 1.9	22.3 $\pm$ 1.1	31.8 $\pm$ 3.0	29.7 $\pm$ 4.6	30.7 $\pm$ 2.7
0.10	23.4 $\pm$ 2.3	19.3 $\pm$ 2.7	21.4 $\pm$ 1.3	34.1 $\pm$ 2.7	32.8 $\pm$ 3.4	33.5 $\pm$ 2.9
0.30	23.7 $\pm$ 4.2	23.4 $\pm$ 3.4	23.6 $\pm$ 3.8	34.9 $\pm$ 3.7	30.2 $\pm$ 3.9	32.5 $\pm$ 2.4
0.50	23.4 $\pm$ 1.1	24.5 $\pm$ 1.9	24.0 $\pm$ 1.4	34.6 $\pm$ 3.0	28.1 $\pm$ 2.5	31.4 $\pm$ 0.7
<i>random-p</i>						
0.02	25.5 $\pm$ 2.0	29.7 $\pm$ 2.2	27.6 $\pm$ 1.2	39.3 $\pm$ 1.5	32.3 $\pm$ 2.7	35.8 $\pm$ 1.0
0.05	29.7 $\pm$ 1.9	33.3 $\pm$ 1.9	31.5 $\pm$ 1.8	43.0 $\pm$ 2.2	38.5 $\pm$ 1.9	40.8 $\pm$ 0.4
0.10	36.2 $\pm$ 1.9	39.6 $\pm$ 0.7	37.9 $\pm$ 0.6	52.9 $\pm$ 2.9	46.9 $\pm$ 3.4	49.9 $\pm$ 2.1
0.30	66.9 $\pm$ 2.7	64.1 $\pm$ 2.5	65.5 $\pm$ 0.4	74.2 $\pm$ 0.6	72.9 $\pm$ 3.9	73.6 $\pm$ 2.3
0.50	80.0 $\pm$ 0.7	73.4 $\pm$ 0.0	76.7 $\pm$ 0.4	87.2 $\pm$ 2.0	81.2 $\pm$ 2.5	84.2 $\pm$ 2.2
sign_only	85.2 $\pm$ 1.7	74.5 $\pm$ 3.7	79.8 $\pm$ 1.4	83.8 $\pm$ 1.3	80.2 $\pm$ 1.9	82.0 $\pm$ 1.1

and §B.3 replicates the subspace asymmetry of §5 on this model/task pair.

B.1 Training and Reproduction Setup

Base model and data. We start from Qwen3-8B (Qwen Team, 2025b). Following Yu et al. (2026), we fix the history length $H = 5$ to deal with more complex scenarios. As τ^2 -Bench provides only 178 training tasks, we roll out 6 trajectories per task – three with GPT-4.1 as the user simulator and three with Qwen3-235B-A22B-Instruct – and convert trajectories into transition triplets. To prevent the model from memorizing database contents, we replace tool-response values with their OpenAPI schemas.

Sim-to-real reward. Following Yu et al. (2026): embedding-based binary similarity for natural-language user responses (Qwen3-Embedding-8B, $\eta = 0.6$), and a discretised ROUGE-L score for structured tool responses.

WMRL stage. 2 epochs, learning rate $1\text{e}-6$, effective batch size 32, GRPO group size 16. We use similar filtering pipeline as Yu et al. (2026). Each checkpoint is trained on two NVIDIA H200 GPUs for around 10 hours. The data size is $\sim 6\text{K}$.

PRL stage. GRPO with terminal task-success reward, $\gamma = 1.0$, max 30 steps per episode, 200 update steps, group size 8. We use Qwen3-235B-

A22B-Instruct as the user simulator throughout. We also use the modified VerI-Agent GitHub repo for training and each checkpoint is trained on four NVIDIA H200 GPUs for around 100 hours.

Table 12: Replicated τ^2 -Bench performance. AVG@3 is mean $\pm$ std success rate over 3 seeds across retail, telecom, and airline; pass@ k (Chen et al., 2021) is the probability that at least one of k rollouts succeeds.

Model	AVG@3	pass@1	pass@3
ReAct (base)	33.7 $\pm$ 2.5	37.0	49.0
WMSFT	26.0 $\pm$ 4.3	30.0	46.0
WMRL	36.3 $\pm$ 0.5	36.0	52.0
PRL	39.3 $\pm$ 2.6	38.0	58.0
WMSFT $\rightarrow$ PRL	38.3 $\pm$ 4.5	41.0	59.0
WMRL $\rightarrow$ PRL	45.0 $\pm$ 1.6	43.0	64.0

B.2 Per-layer Effective Rank

We reproduce the main-text per-layer effective-rank plot (Fig. 7) on Qwen3-8B + τ^2 -Bench checkpoints in Fig. 10, using the same plot format (one line per checkpoint, averaged across all target modules). The qualitative ordering is preserved across all 36 layers: WMRL sits visibly above the RL-stage checkpoints (PRL, WMRL $\rightarrow$ PRL, WMSFT $\rightarrow$ PRL, clustered in the middle), and WMSFT trails them in the deeper layers – matching the Qwen2.5/ALFWorld pattern on a different model and benchmark.

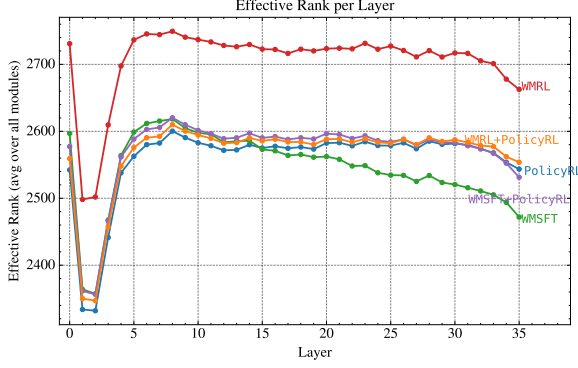


Figure 10: **Per-layer eRank(Δ) of post-training on Qwen3-8B + τ^2 -Bench**, averaged across all target modules. Same plot format as the main-text Fig. 7: WMRL (red) keeps the highest per-layer rank; the RL-after-WM variants (WMRL $\rightarrow$ PRL, WMSFT $\rightarrow$ PRL) and PRL cluster closely in the middle; WMSFT (green) is comparable in the early layers but drifts below the RL-trained checkpoints in the deeper layers.

B.3 Cross-model Replication of the Subspace Asymmetry

The main-text geometric analyses in §5 are conducted on Qwen2.5-7B-Instruct fine-tuned on ALFWorld. To check that the U-disjoint / V-shared asymmetry is not an artefact of a single model or task, we replicate the module-wise subspace overlap measurement on a different model and benchmark: Qwen3-8B fine-tuned on τ^2 -Bench. The fine-tuning recipe follows the same two-stage pipeline (WMRL followed by PRL), so we can form the same Δ_{PRL} and $\Delta\Delta_{\text{PRL}}$ updates and compute their leading- k left and right subspace overlap module by module, exactly as in §5.4.

Headline numbers (Tab. 13). The asymmetry holds, and is in fact slightly sharper on Qwen3. Mean rank-1 *left* overlap is 0.04 on both models (essentially identical), while mean rank-1 *right* overlap is 0.81 on Qwen3 versus 0.63 on Qwen2.5. The Frobenius cosine between Δ_{PRL} and $\Delta\Delta_{\text{PRL}}$ is 0.015 on Qwen3 (vs. 0.028 on Qwen2.5), confirming that the two updates are close to orthogonal as matrices on both models.

Module-wise heatmaps. Fig. 11 reproduces the rank-1 left/right overlap heatmaps of §5.4 on Qwen3. Visually the pattern is the same as on Qwen2.5 (cf. Fig. 4): the left heatmap is uniformly dark (near-zero overlap on output directions) while the right heatmap is uniformly bright (high overlap on input-feature directions). A small handful of early-FFN modules (up_proj/gate_proj in layers

Table 13: Cross-model replication of the module-wise subspace overlap (Δ_{PRL} vs. $\Delta\Delta_{\text{PRL}}$) and Frobenius cosine, averaged over all target modules in each model.

metric	Qwen2.5-7B (ALFWorld)	Qwen3-8B (τ^2 -Bench)
n_modules	196	252
Frobenius cosine, mean	0.028	0.015
<i>Left subspace overlap</i>		
$k=1$, mean	0.036	0.043
$k=50$, mean	0.158	0.157
$k=250$, mean	0.239	0.238
<i>Right subspace overlap</i>		
$k=1$, mean	0.635	0.811
$k=50$, mean	0.411	0.508
$k=250$, mean	0.393	0.416

1–4) show the reverse pattern – shared on the left, disjoint on the right – which we note but do not interpret further here; outside this small block, the U-disjoint / V-shared geometry holds module-wise across all 36 layers.

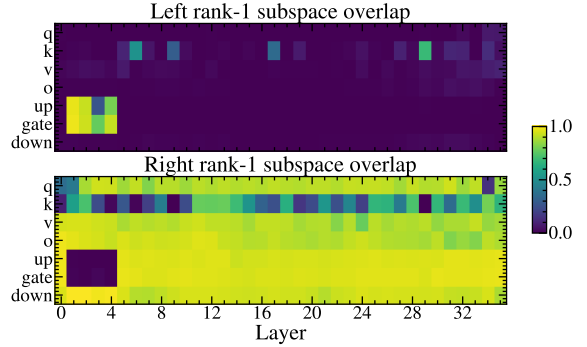


Figure 11: **Module-wise rank-1 subspace overlap between Δ_{PRL} and $\Delta\Delta_{\text{PRL}}$ on Qwen3-8B + τ^2 -Bench.** The top panel shows the left-side overlap ($k=1$, mean 0.04), nearly orthogonal, while the bottom panel shows the right-side overlap ($k=1$, mean 0.81), highly aligned at the dominant input direction – matching the Qwen2.5/ALFWorld pattern.

Does the geometric asymmetry translate to a behavioral asymmetry? To check whether the U-disjoint / V-shared geometry reproduces *behaviorally* on Qwen3, we run the same parallel/orthogonal projection sweep that produced the main-text left/right curves on the Qwen3 + τ^2 -Bench checkpoints, and then we compare the $k=1$ and $k=250$ rows against the Qwen2.5/ALFWorld numbers in Tab. 14.

Geometry replicates The qualitative direction is preserved: on Qwen3, orthogonal-to- Δ_{WMRL} projec-

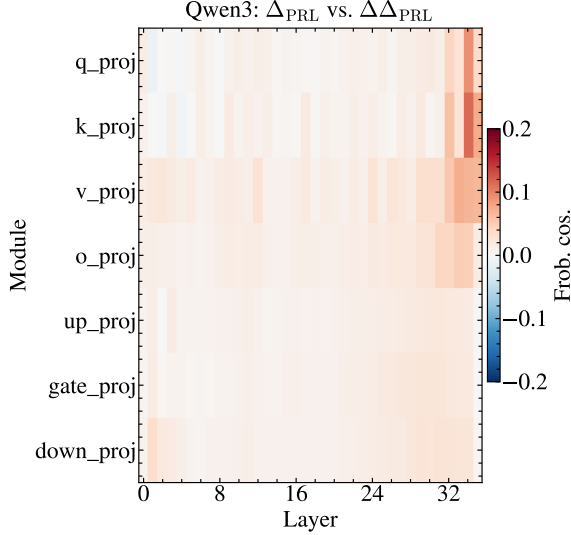


Figure 12: Per-module Frobenius cosine similarity between Δ_{PRL} and $\Delta\Delta_{\text{PRL}}$ on Qwen3-8B + τ^2 -Bench. The vast majority of modules are at or near zero; only a small subset of attention projections in the last three layers shows mild positive similarity ($\lesssim 0.10$). Module-wise mean is 0.015.

tions of Δ_{PRL} at $k=1$ are still higher than parallel ones (40.3 vs. 34.0, gap +6.3), and the parallel projection sits at the base+ Δ_{WMRL} level rather than near the oracle – the sign of the asymmetry matches Qwen2.5. What is different is the *magnitude*: on Qwen2.5 the $k=1$ orth–par gap is +50 points, on Qwen3 it is +6. Three factors might explain most of this attenuation:

- *Smaller dynamic range for Δ_{PRL} .* On Qwen2.5, Δ_{PRL} buys +60.6 points over base (17.7 $\rightarrow$ 78.3); on Qwen3 it buys only +5.6 (33.7 $\rightarrow$ 39.3). The behavioral signal that any projection of Δ_{PRL} can corrupt or preserve is proportionally smaller.
- *Smaller two-stage gain over Δ_{PRL} .* The $\Delta_{\text{WMRL}} + \Delta\Delta_{\text{PRL}}$ oracle is only +5.7 above Δ_{PRL} alone on Qwen3 (vs. +6.1 on Qwen2.5 in absolute, but starting from a far higher base) – and as a fraction of headroom this is much thinner. Recovering even 80% of that gain is a ~ 4.6 point swing that competes with per-repeat noise of ± 1 –4 points.
- *Lower absolute success rates and a harder task.* τ^2 -Bench operates in the 30–45% band where multi-tool long-horizon failures dominate; small per-direction perturbations do not change which tasks succeed in nearly as many cases as on ALFWorld’s 17–88% range.

Table 14: Cross-model behavioral projection results. Numbers are end-to-end success rate (%) averaged over 3 repeats. Same projection sweep as Tabs. 6 and 8, restricted to $k=1$ and $k=250$. Geometric asymmetry replicates cleanly on Qwen3 (Tab. 13), but the behavioral separation between parallel and orthogonal projections is much smaller, for reasons we discuss below.

	Qwen2.5-7B (ALFWorld)	Qwen3-8B (τ^2 -Bench)
base	17.7	33.7
Δ_{WMRL} alone	32.7	36.3
Δ_{PRL} alone	78.3	39.3
$\Delta_{\text{WMRL}} + \Delta\Delta_{\text{PRL}}$ (oracle)	84.4	45.0
<i>Left-space proj. of Δ_{PRL} onto Δ_{WMRL}</i>		
parallel, $k=1$	32.5	34.0
parallel, $k=250$	43.9	35.0
orthogonal, $k=1$	82.5	40.3
orthogonal, $k=250$	84.4	40.7
<i>Right-space proj. of Δ_{PRL} onto Δ_{WMRL}</i>		
parallel, $k=1$	42.3	35.3
parallel, $k=250$	65.6	39.0
orthogonal, $k=1$	83.3	39.7
orthogonal, $k=250$	67.3	40.7

The right-side picture is the weakest: on Qwen2.5, the parallel-on-right projection at $k=250$ recovered 71.8% of the oracle gain over base, evidence that Δ_{PRL} ’s input-feature scope is largely shared with Δ_{WMRL} . On Qwen3, the analogous parallel-on-right projection recovers only 46.9% of the (much smaller) oracle gain – consistent with the above noise/dynamic-range argument rather than with a failure of the geometric claim, since the underlying right-overlap mean is in fact higher on Qwen3 (0.81 vs 0.63).

Takeaway. The U-disjoint / V-shared subspace asymmetry between Δ_{PRL} and $\Delta\Delta_{\text{PRL}}$, and the near-zero Frobenius cosine, are reproduced on a different base model (Qwen3-8B vs Qwen2.5-7B), a different post-training corpus (τ^2 -Bench vs ALFWorld), and a different model depth (36 vs 28 layers), with essentially identical left numbers and an even sharper right alignment. The behavioral projection sweep on Qwen3 carries the same qualitative sign (orth > par on the left at low k , no collapse under either projection), but its margins are compressed into the noise band by the smaller absolute Δ_{PRL} contribution and the harder benchmark. Read together, these two appendices indicate that the geometric complementarity diagnosed in §5 may be a property of the two-stage post-training recipe rather than of a particular model/task, while the strength with which it shows up in end-to-end

success requires a task and model where Δ_{PRL} itself has substantial headroom to begin with. However, we caution against over-interpreting these results: these experiments cannot establish generality across models and agent types.

C Preservation Method Details

We study two ways to preserve the world-modeling ability acquired during WMRL while adapting the policy with PRL. The first is *training-free*: merge two fine-tuned checkpoints in weight space. The second is *training-based*: add an online world-modeling SFT loss during PRL. In all cases, the preservation method produces a single deployable policy checkpoint.

Training-free checkpoint merging. The merging methods take the base model M_0 and two task checkpoints as input: always M_{WMRL} for task A , and either M_{PRL} or the sequential checkpoint $M_{\text{WMRL}+\text{PRL}}$ (WMRL $\rightarrow$ PRL) for task B . For each target weight matrix $W^{(k)}$, we form task vectors

$$\tau_i^{(k)} = W_i^{(k)} - W_0^{(k)}, \quad i \in \{A, B\}.$$

We merge only the seven projection matrices in each transformer layer,

q, k, v, o, up, gate, down,

and leave embeddings, layer norms, and the lm-head at their base-model values.

Plain TIES. Plain TIES operates directly on the raw task vectors τ_A, τ_B flattened from the entire model’s parameters. For each pair of models, we stack the two task vectors, keep the top- $K\%$ entries by magnitude ($K=40$), resolve sign conflicts with `sum_of_values`, and average the surviving entries using a disjoint mean, i.e. each coordinate is divided by the number of non-zero contributors rather than by the total number of tasks. The merged checkpoint is

$$W_{\text{merged}} = W_0 + \text{TIES}(\tau_A, \tau_B).$$

KnOTS. KnOTS first represents the two module-wise task vectors in a shared low-rank SVD frame, then applies the same TIES masking and aggregation in that frame. In the default U -sharing version, the two task matrices are concatenated along the input dimension:

$$\begin{bmatrix} \tau_A^{(k)} & \tau_B^{(k)} \end{bmatrix} = U^{(k)} \Sigma^{(k)} V^{(k)\top}.$$

Algorithm 1 Online WMSFT during PRL

Require: Policy π_θ , downsampling ratio $\rho = 0.2$, WM-SFT weight $\lambda = 1.0$

- 1: **for** each PRL update step **do**
 - 2: Collect on-policy rollouts with π_θ
 - 3: Extract real WM-SFT pairs $\mathcal{D}_{\text{WM}} = \{((s_{t-H:t}, a_t), s_{t+1})\}$
 - 4: Downsample no-op transitions ($s_{t+1} = s_t$) to at most ρ times the number of changed-observation transitions
 - 5: Randomly keep 32 samples each step
 - 6: Compute $\mathcal{L}_{\text{GRPO}}$ on the rollout batch
 - 7: Compute $\mathcal{L}_{\text{WM-SFT}}$ by predicting s_{t+1} from $(s_{t-H:t}, a_t)$
 - 8: Update θ using $\mathcal{L}_{\text{GRPO}} + \lambda \mathcal{L}_{\text{WM-SFT}}$
 - 9: **end for**
 - 10: **return** Updated policy π_θ
-

This gives a shared left basis $U^{(k)}$, corresponding to a shared output subspace. The per-task right factors are split from $\Sigma V^\top$, TIES is applied to those factors, and the merged delta is reconstructed as

$$\Delta^{(k)} = U^{(k)} V_{\text{merged}}^{(k)}.$$

We also evaluate a V -sharing variant, motivated by the stronger right-subspace overlap observed in §5. This variant transposes each task matrix before the joint SVD, so that the shared basis lies on the input side rather than the output side. After merging in this transposed frame, the reconstructed matrix is transposed back to the original (*out*, *in*) shape. We modify KnOTS’s official repository² to enable full parameter update merging.

For both KnOTS variants, we use the same TIES hyperparameters as plain TIES: $K=40$, `sum_of_values` sign resolution, disjoint-mean aggregation, scaling coefficient 1.0, and no DARE pre-processing. Singular directions with negligible singular values are discarded.

Training-based: online WM-SFT during PRL.

We follow the PRL GRPO training setup of Yu et al. (2026), but add an auxiliary online world-modeling SFT update. At each training step, we reuse the same on-policy rollouts collected for GRPO and convert each realized transition into a next-observation prediction example: given the recent observation history $s_{t-H:t}$ and the executed action a_t , the model predicts the environment’s

²<https://github.com/gstoica27/knots>

next observation s_{t+1} . Hence, the WM signal is “free” in the sense of the concurrent ECHO recipe of Shrivastava et al. (2026): it requires no additional environment interaction. The training objective is

$$\mathcal{L} = \mathcal{L}_{\text{GRPO}} + \lambda \mathcal{L}_{\text{WM-SFT}}, \quad \lambda = 1.0$$

with all other PRL hyperparameters unchanged.

Table 15: KnOTS-TIES preservation interventions on Qwen3-8B + τ^2 -Bench. We compare the source checkpoints with training-free KnOTS-TIES merges at top- $K=40$, using either the default shared U basis or our shared V -basis variant. The two merge blocks differ in the second task arm: PRL from base, or the trained sequential WMRL $\rightarrow$ PRL checkpoint. **Bold** = best within each sub-block.

Method	AVG@3	pass@1	pass@3
PRL	39.3 ± 2.6	38.0	58.0
WMRL $\rightarrow$ PRL	45.0 ± 1.6	43.0	64.0
<i>training free: WMRL + PRL</i>			
KnOTS-TIES (U)	38.0 ± 0.8	37.0	54.0
KnOTS-TIES (V)	40.7 ± 2.9	37.0	58.0
<i>training free: WMRL + (WMRL $\rightarrow$ PRL)</i>			
KnOTS-TIES (U)	38.0 ± 3.6	35.0	56.0
KnOTS-TIES (V)	39.3 ± 4.2	41.0	57.0

KnOTS replication on τ^2 -Bench. We replicate the training-free KnOTS-TIES merges from §C on Qwen3-8B + τ^2 -Bench, using the same top- $K=40$, `sum_of_values` sign resolution, and `merging_type=mean` as in the ALFWorld experiments (Table 15). The results broadly favor the shared right/input singular basis (V -sharing) over the default U -sharing variant, consistent with the input-side geometry suggested by §5. In both merge settings, the V -sharing variant attains the best AVG@3 within the training-free block, and in the WMRL + (WMRL $\rightarrow$ PRL) setting it also improves pass@1 and pass@3. However, the best training-free merge still remains below the trained WMRL $\rightarrow$ PRL sequential baseline, suggesting that simple checkpoint merging is not sufficient to recover the full benefit of staged training on this more complex benchmark. For cost reasons, we do not replicate the online WMSFT auxiliary-loss baseline on τ^2 -Bench.

D WMRL Data Example

We show one representative training example from ALFWorld and one from τ^2 -Bench. Each example consists of the model prompt, and the ground-truth

next observation used as the supervision target. Part of the content is omitted for display.

For WMSFT and entropy probing in Figure 6, we revise the final instruction to disable thinking.

ALFWorld example

Prompt

You are an expert agent operating in the ALFRED Embodied Environment. Your task is to: look at pillow under the desk lamp. Prior to this step, you have already taken 7 step(s). Below are the most recent 2 observations and the corresponding actions you took: [Observation 6: 'You arrive at shelf 1. On the shelf 1, you see a pencil 2, and a pencil 1.', Action 6: '[<action>go to desk 1</action>']' [Observation 7: 'Nothing happens.', Action 7: '<action>go to desk 1</action>']']

You are now at step 8 and your current observation is: Nothing happens.

Your admissible actions of the current situation are: ['examine shelf 1' 'go to bed 1' 'go to desk 1' 'go to desk 2' 'go to garbagecan 1' 'go to shelf 2' 'go to shelf 3' 'go to shelf 4' 'go to shelf 5' 'inventory' 'look' 'take pencil 1 from shelf 1' 'take pencil 2 from shelf 1'].

Potential action: <action>go to desk 1</action>

Now, your task is to predict the immediate next observation after executing the potential action above. You should first **briefly** reflect on the previous steps and current situation. Then think about what the next observation will look like after taking this action, and also whether the task is completed after taking this action. This reflection and reasoning process MUST be enclosed within <think> </think> tags. Once you've finished your reasoning, present both your final prediction of the next observation (use the past and current observations as examples!) and task completion status within <next_state> </next_state> tags.

Ground-truth next observation

You arrive at desk 1. On the desk 1, you see a creditcard 1, and a keychain 1. Task is not yet completed.

ALFWorld Non-thinking Prompt

...
Now, your task is to predict the immediate next observation after executing the potential action above.
Directly present your final prediction of the next observation. DO NOT generate anything else.

Ground-truth next observation

Hmm, I don't have the order ID on me right now -- everything's kind of scattered. Is there another way we can find it? Maybe by the date I ordered or the tablet model? I think it was within the last two weeks.

Task is not yet completed.

Tau2-Bench Non-thinking Prompt

...
DO NOT perform any thinking. Directly present your final prediction of the next user/tool response and task completion status.

τ^2 -Bench example

Prompt, abbreviated

```
<instructions>
You are a customer service agent that helps
the user according to
the <policy> provided below.
...
</instructions>

<policy>
# Retail agent policy
[... full retail policy elided: ~6k chars
covering cancel/modify
pending orders, return/exchange delivered
orders, identity
authentication rules, payment-method rules,
etc. ...]
</policy>

<tools>
[... ~30 tool schemas elided:
find_user_id_by_email,
find_user_id_by_name_zip, get_user_details,
get_order_details,
return_delivered_order_items,
exchange_delivered_order_items,
cancel_pending_order,
modify_pending_order_items,
transfer_to_human,
...]
</tools>

# Dialogue so far
user: Hi! I need to return two tablets I
bought recently. Can you
help me with that?
assistant: I can help you with that. To
process your return, I'll
need to verify your identity first.
...
potential assistant response: I've located
your account, Chen Silva
(user ID: chen_silva_7485). To process your
return, I'll need the
order ID of the tablets you purchased.

Once you've finished your thinking, format
both your final prediction of the next
user/tool response and task completion status
within <next_state> </next_state> tags.
```

E WM-SFT Prompt Format

We use a lightweight next-observation prediction template for online WMSFT to mitigate the collapse in the separate training stage. The prompt contains the previous raw environment observation and the agent's extracted action; the supervision target is only the next environment observation.

Template

Given the previous observation and the action, predict the next observation.

Previous observation:
{prev_obs}

Action:
{action}

Next observation:

Concrete instance

<|im_start|>user
Given the previous observation and the action, predict the next observation.

Previous observation:
Nothing happens.

Action:
go to desk 1

Next observation:<|im_end|>
<|im_start|>assistant

Only these target tokens have loss_mask=1

Supervision target

You arrive at desk 1. On the desk 1, you see a creditcard 1, and a keychain 1.<|im_end|>